0.0.1 Stochastic Process

A stochastic or random process, denoted by $\{s_t\}_{t \in T}$, can be defined as a collection of random variables that is indexed by some mathematical set T

- Index set T has the interpretation of time.
- The set T is typically $\mathbb{N}$ or $\mathbb{R}$.

0.1 Markov chain

0.1.1 State transition probability

For a markov state s and a successor state s', the state transition probability is defined as

$$\mathcal{P}_{ss'} = \Pr\{s_{t+1} = s' | s_t = s\}$$

We define the state transition probability matrix as

$$\mathcal{P} = \left[\mathcal{P}_{ss'}\right]$$

0.2 Markov Reward Process (MRP)

A Markov Reward Process (MRP) is a Markov chain with an additional reward function. It is defined by the tuple $\langle \mathcal{S}, \mathcal{P}, \mathcal{R}, \gamma \rangle$, where:

- $\mathcal{S}$ is the state space.
- $\mathcal{P}$ is the state transition probability matrix.
- $\mathcal{R}$ is the reward function, with $\mathcal{R}(s_t) = r_{t+1}$ being the reward for being in state s_t at time t.
- $\gamma \in [0, 1)$ is the discount factor, which determines the importance of future rewards.

0.3 Policy evaluation

0.3.1 Value function with respect to a policy

We define $V^\pi(s)$ as the expected return (cumulative future reward) when starting from state s and following policy π:

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} | s_t = s \right]$$

0.3.2 Decomposition of state value function

The state value function can be decomposed into the immediate reward and the expected value of the next state,

$$V^\pi(s) = \mathbb{E}_\pi \left[r_{t+1} + \gamma V^\pi(s_{t+1}) | s_t = s \right]$$

Expanding the expectation, with $\mathcal{R}^a_{ss'} = \mathcal{R}(s, a, s')$, we have

$$\mathbb{E}_\pi \left[r_{t+1} | s_t = s\right] = \sum_a \pi(a|s) \sum_{s'} \mathcal{P}^a_{ss'} \mathcal{R}^a_{ss'}$$
$$\mathbb{E}_\pi \left[\gamma V^\pi(s_{t+1}) | s_t = s\right] = \sum_a \pi(a|s) \sum_{s'} \mathcal{P}^a_{ss'} V^\pi(s')$$
$$\implies V^\pi(s) = \sum_a \pi(a|s) \sum_{s'} \mathcal{P}^a_{ss'} \left(\mathcal{R}^a_{ss'} + \gamma V^\pi(s')\right)$$

The above is called the **Bellman Evaluation operator**.

0.3.3 Matrix formulation of Bellman Evaluation equation

We define,

$$\mathcal{P}^\pi(s'|s) := \sum_{a \in \mathcal{A}} \pi(a|s) \mathcal{P}^a_{ss'}$$
$$\mathcal{R}^\pi(s) := \sum_{a \in \mathcal{A}} \pi(a|s) \sum_{s'} \mathcal{P}^a_{ss'} \mathcal{R}^a_{ss'} = \mathbb{E}(r_{t+1} | s_t = s)$$

Using the above definitions, for finite state MDP, we can rewrite the Bellman Evaluation equation in matrix form as

$$V^\pi = \mathcal{R}^\pi + \gamma \mathcal{P}^\pi V^\pi$$

or in explicit form,

$$V^\pi = \left(I - \gamma \mathcal{P}^\pi\right)^{-1} \mathcal{R}^\pi$$

0.4 Relation between Markov Decision Process and Markov Reward Process

MDP + Policy = MRP

0.5 Optimal policy

We define a partial ordering over policies,

$$\pi \geq \pi', \iff V^\pi(s) \geq V^{\pi'}(s) \quad \forall s \in \mathcal{S}.$$

0.5.1 Theorem: Existence of optimal policy

For any MDP, there exists an optimal policy π_* such that

$$V^{\pi_*}(s) \geq V^\pi(s) \quad \forall \pi \in \Pi, \forall s \in \mathcal{S}.$$

and all optimal policies achieve the same value function, i.e.,

$$V_*(s) = V^{\pi'_*}(s) \quad \forall \pi'_* \in \Pi_*.$$

0.6 Action value function

We define the action value function $Q^\pi(s,a)$ as the expected return when starting from state s, taking action a, and then following policy π:

$$Q^\pi(s,a) = \mathbb{E}_\pi\left[\sum_{k=0}^{\infty}\gamma^k r_{t+k+1}|s_t = s, a_t = a\right]$$

We can decompose the action value function as follows

$$Q^\pi(s,a) = \mathbb{E}_\pi\left[r_{t+1} + \gamma V^\pi(s_{t+1})|s_t = s, a_t = a\right]$$

Expanding the expectation, we have

$$Q^\pi(s,a) = \sum_{s'}\mathcal{P}^a_{ss'}\left(\mathcal{R}^a_{ss'} + \gamma\sum_{a'}\pi(a'|s')Q^\pi(s',a')\right)$$

0.7 Relationship between state ($V^\pi(\cdot)$) and action ($Q^\pi(\cdot,\cdot)$) value functions

The relationship between the state value function and the action value function is given by

$$V^\pi(s) = \sum_{a\in\mathcal{A}}\pi(a|s)Q^\pi(s,a)$$

and

$$Q^\pi(s,a) = \sum_{s'\in\mathcal{S}}\mathcal{P}^a_{ss'}\left(\mathcal{R}^a_{ss'} + \gamma V^\pi(s')\right)$$

As seen previously, all optimal policies achieve the optimal action value function,

$$Q_*(s,a) = \max_{\pi\in\Pi} Q^\pi(s,a)$$

and

$$Q_*(s,a) = Q^{\pi_*}(s,a) \quad \forall\pi_* \in \Pi_*$$

0.8 Greedy policy

For a given $Q^\pi(\cdot,\cdot)$, define $\pi'(s)$ as,

$$\pi'(s) = \text{greedy}(Q) = \begin{cases} 1 & \text{if } a = \arg\max_{a'\in\mathcal{A}} Q^\pi(s,a') \\ 0 & \text{otherwise}\end{cases}$$

For a given $V^\pi(\cdot)$, define $\pi'(s)$ as,

$$\pi'(s) = \text{greedy}(V) = \begin{cases} 1 & \text{if } a = \arg\max_{a\in\mathcal{A}}\left[\sum_{s'\in\mathcal{S}}\mathcal{P}^a_{ss'}\left(\mathcal{R}^a_{ss'} + \gamma V^\pi(s')\right)\right] \\ 0 & \text{otherwise}\end{cases}$$

0.9 Relationship between optimal state ($V_*(\cdot)$) and optimal action ($Q_*(\cdot,\cdot)$) value functions

We have

$$V_*(s) = \max_{a\in\mathcal{A}} Q_*(s,a)$$
$$Q_*(s,a) = \sum_{s'\in\mathcal{S}} \mathcal{P}^a_{ss'}\left(\mathcal{R}^a_{ss'} + \gamma V_*(s')\right)$$

0.10 Policy Iteration

We have

$$\pi_0 \xrightarrow{E} v_{\pi_0} \xrightarrow{I} \pi_1 \xrightarrow{E} \dots \xrightarrow{I} \pi_* \xrightarrow{E} v_*,$$

where E is **Policy Evaluation** and I is **Policy Improvement**.

The full algorithm is as follows

Algorithm: Policy Iteration

Input: An MDP $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ and an initial policy π_0
Output: An optimal policy π^*

1. **Initialization**
   Set $k \leftarrow 0$

2. **Repeat**
   a. *(Policy Evaluation)*
      Compute the value function of the current policy π_k by solving the Bellman expectation equation for all $s \in \mathcal{S}$:
      $V^{\pi_k}(s) \leftarrow \sum_{a\in\mathcal{A}} \pi_k(a|s) \sum_{s'\in\mathcal{S}} P^a_{ss'}\ [R^a_{ss'} + \gamma V^{\pi_k}(s')]$

   b. *(Policy Improvement)*
      Improve the policy by acting greedily with respect to V^{π_k} for all $s \in \mathcal{S}$:
      $\pi_{k+1}(s) \leftarrow \arg\max_{a\in\mathcal{A}} \left(\sum_{s'\in\mathcal{S}} P^a_{ss'}[R^a_{ss'} + \gamma V^{\pi_k}(s')]\right)$

   c. Set $k \leftarrow k+1$

   Until $\pi_k(s) = \pi_{k-1}(s)$ for all $s \in \mathcal{S}$ (policy is stable)

3. **Return** $\pi^* \leftarrow \pi_k$

0.11 Value Iteration

Algorithm: Value Iteration

Input: An MDP $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ and an value function V_0
Output: An optimal policy π^*

1. **Initialization**
   Set $k \leftarrow 0$

2. **Repeat**
   a. for $s \in \mathcal{S}$
      $V_{k+1}(s) \leftarrow \max_{a \in \mathcal{A}} \left[\sum_{s' \in \mathcal{S}} \mathcal{P}^a_{ss'} \left(\mathcal{R}^a_{ss'} + \gamma V_k(s') \right) \right]$

   b. Set $k \leftarrow k + 1$

   Until $V_k(s) = V_{k-1}(s)$ for all $s \in \mathcal{S}$ (value function is stable)

3. **Return** $\pi^* \leftarrow \arg\max_{a \in \mathcal{A}} \left[\sum_{s' \in \mathcal{S}} \mathcal{P}^a_{ss'} \left(\mathcal{R}^a_{ss'} + \gamma V_k(s') \right) \right]$

0.11.1 Value function space

Let

$$\mathcal{V} = \{ f : \mathcal{S} \to \mathbb{R} \mid \sup_{s \in \mathcal{S}} |f(s)| < \infty \},$$

be the space of all value functions.

0.11.2 Bellman Evaluation Operator

We define Bellman Evaluation Operator $\mathcal{L}^\pi : \mathcal{V} \to \mathcal{V}$ as,

$$\mathcal{L}^\pi(v) = \mathcal{R}^\pi + \gamma \mathcal{P}^\pi v$$

0.11.3 Bellman Optimality Operator

We define Bellman Optimality Operator $\mathcal{L} : \mathcal{V} \to \mathcal{V}$ as,

$$\mathcal{L}(v) = \max_{a \in \mathcal{A}} \left[\mathcal{R}^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}^a_{ss'} v \right]$$

0.12 Prediction and Control in Reinforcement Learning

1 Pictures

2 Function approximation methods

2.1 Monte Carlo based value function fitting

Use Monte Carlo to get rollouts

Perform supervised regression

$$y_i = \sum_{H}^{k=0} \left(\gamma^k r_{t+k+1} \mid s_t = s \right)$$

$$(s_i, y_i)$$

$$L(\phi) = \frac{1}{2} \sum_{N}^{i=1} \left(V_\phi^\pi(s_i) - y_i \right)^2$$

2.1.1 See also

2.1.2 References